

Regulating relative prices: mandated discounts and price menus^{*}

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Abstract

Brazil's half-price law requires cinemas to charge eligible consumers half the full price, while leaving the firm free to set the full price. This paper presents a model of how this relative price regulation reshapes the firm's price menu and its welfare consequences. When the mandated discount is more generous than the firm would voluntarily offer, the firm raises the full price, and this response is stronger when more consumers are eligible for the discount. Firms may also offer other promotions. With independent promotional pricing, the mandate leaves that price unchanged, but the higher full price makes the promotion appear more generous. This regulation may increase attendance while reducing welfare. It attracts additional eligible consumers, but it deters full-price consumers, whose purchases generate higher markups. A cap set at the same eligible price prevents the increase in the full price, thus eliminating the menu distortion and raising profit and consumer surplus relative to the mandate.

Keywords: price discrimination, relative prices, mandated discounts, price menus

JEL classification: L11, L82, Z11, D42, D21, D22

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1 Introduction

Price regulation is often analysed as a constraint on a single price, but firms usually choose prices as part of a broader menu. Thus, a regulation that fixes the ratio between two prices in such a menu generates a different problem. Such a rule leaves a firm free to choose the overall level of the linked prices and any other prices in its menu, but those choices must accommodate the regulated relationship. Hence, the incidence of the regulation depends on how the firm adjusts its whole menu. In this paper, I study the pricing and welfare consequences of this form of relative price regulation.

An example of this type of regulation is Brazil's half-price law. It requires cinemas, and other firms in other cultural industries, to sell tickets to students and other eligible groups at exactly half the posted full price, while leaving cinemas free to choose that full price and to offer any other voluntary promotion. A related policy exists in Spain, where residents of the Canary Islands, Balearic Islands, Ceuta, and Melilla receive a 75% discount on airfares, while airlines choose the fare on which the discount is calculated (Calzada and Fageda 2012; Fageda et al. 2016, 2017).¹ In both cases, the benefit received by eligible consumers is defined relative to a price chosen by the firm.

Motivated by this class of policies, I build a monopoly model of relative price regulation.² When the mandated discount is more generous than the firm would voluntarily offer, the firm cannot retain the prices it would independently choose for each consumer group. It then raises the full price, partly offsetting the mandated discount, while the eligible price remains below its voluntary level. A larger share of eligible consumers strengthens the incentive to raise the full price. Consequently, consumers who pay the full price bear part of the adjustment due to the regulation.

I then extend the model by adding a voluntary promotional ticket.³ When the pro-

¹The government's rationale for this discount is to ease integration between these territories and the main Spanish territory in the Iberian Peninsula.

²This is motivated by the Brazilian cinema industry. Approximately 71% of Brazilian municipalities with an active cinema have a single cinema complex.

³Administrative data from the Brazilian National Film Agency show that the share of promotional tickets increased from 17.6% in 2018 to 24.2% in 2025.

motion serves a separate consumer group and the firm can choose its effective price independently, the regulation leaves the promotional price unchanged. As the posted full price rises, however, the firm increases the advertised promotional discount to preserve that price. A larger advertised discount can therefore arise without any reduction in the promotional price paid by consumers.

To analyse the welfare effects of this regulation, I compare the regulated menu with voluntary third-degree price discrimination. The lower price for the eligible group attracts additional consumers, while the higher full price deters others. The welfare effect depends on the surplus generated by the additional eligible admissions relative to the surplus lost from the deterred full price admissions. Thus, total attendance can rise while welfare falls. With a general demand function, I show how demand curvature and monopoly markups determine the local welfare and output effects of moving away from the firm's voluntary price ratio. Under linear demand and a common positive marginal cost for both groups, I characterise these effects over the range of prices in which both groups remain served. A mandate slightly more generous than the firm's voluntary discount can increase attendance while reducing welfare. Under certain conditions, making the discount sufficiently more generous reverses the welfare effect.

If the government objective is to allow resource constrained consumers to access to cultural activities, an alternative policy would be to cap the price to eligible consumers. I then compare the ratio mandate with a cap set at the same eligible price. When the mandated discount is more generous than the firm's voluntary discount, both policies give eligible consumers the same price. However, the cap removes the link between the eligible and full prices, allowing the firm to restore the full price to its voluntary level. As a consequence, consumers facing the full price pay less, while eligible consumers face no change. The firm also earns more because it can choose its preferred full price. The matched cap therefore raises profit, consumer surplus, and total welfare relative to the ratio mandate.

Related literature. This paper is directly related to the literature on the regulation of a firm's price structure. Price caps and average revenue constraints leave firms

with some flexibility over individual prices but can distort relative prices within the regulated basket (Bradley and Price 1988; Armstrong et al. 1994; Cowan 1997a,b). Armstrong and Vickers (1991) study price discrimination under an average price constraint, while Cowan (2018) studies regulations that equalise profit margins or price-cost ratios across markets. In the present paper, I study a regulator that fixes the ratio between two consumer prices and allows the firm to choose their common level. This imposes a different constraint in the firm's decision problem, and its incidence depends on how the firm adjusts the rest of its price menu. I characterise that adjustment and identify conditions under which the regulation increases total demand while reducing welfare.

This paper also contributes on the welfare analysis of third-degree price discrimination (Schmalensee 1981; Varian 1985; Aguirre et al. 2010; Cowan 2012, 2016). This literature primarily compares voluntary price discrimination with uniform pricing. I instead consider a price ratio chosen by a regulator, with uniform pricing as a special case in which the ratio equals one and voluntary third-degree price discrimination as the benchmark ratio preferred by the firm. For a general class of demand functions, I show how demand curvature and monopoly markups determine the local output and welfare effects of moving away from the voluntary ratio. Under linear demand, I characterise the comparison for larger departures and show that welfare can vary non-monotonically with the generosity of the mandated discount.

The regulation studied in this paper also connects to policies and contractual restrictions that transmit distortions across linked prices. For example, Medicaid's pricing rules tie public drug payments to private sector prices, while Spanish airfare subsidies base reimbursement on firm-chosen fares (Scott Morton 1997; Duggan and Scott Morton 2006; Calzada and Fageda 2012; Fageda et al. 2016, 2017). Platform parity clauses similarly restrict a seller's prices across channels (Boik and Corts 2016; Wang and Wright 2020). Brazil's half-price law differs from these arrangements because it fixes an exact ratio between prices paid by consumer groups within the same channel, without government purchase or reimbursement.

Finally, the policy that motivates this paper relates to research on pricing and price

discrimination in cultural markets. [de Roos and McKenzie \(2014\)](#) study voluntary cinema discounts, while [Leslie \(2004\)](#) and [Courty and Pagliero \(2012\)](#) examine price discrimination in theatre and concerts. [Davis \(2006\)](#) and [Einav \(2007\)](#) study spatial pricing and demand in film exhibition. For Brazil, [Wink Junior et al. \(2016\)](#) shows how half-price laws increase students' consumption of cultural goods. This paper complements that work by studying how a statutory price ratio changes the firm's optimal price menu and the welfare consequences of that adjustment.

The rest of the paper is organised as follows. Section 2 describes Brazil's half-price law and the observed price menu in the cinema industry. Section 3 develops the model and derives the pricing, welfare, and policy comparison results. Section 4 concludes. Proofs are in the appendix.

2 Institution, data, and stylised facts

2.1 Brazil's half-price law

Brazil has a long history of regional laws establishing a minimum discount for cultural and sporting events, beginning in 1990. A national law standardising these rules was approved in 2013. Law No. 12,933/2013 guarantees students admission at half the ticket price effectively charged to the general public and extends the benefit to people with disabilities and, when necessary, their companions, as well as low-income people aged 15-29.⁴ People aged 60 or above have a separate entitlement under the Elderly Citizen Statute, which specifies a discount of at least 50%.⁵

The federal legislation followed exhibitors' concerns about the expansion of half-price ticket sales and fraudulent student identification. According to the National Federation of Film Exhibition Companies, the share of half-price tickets increased from around 40% of cinema admissions in 2001 to 70% in 2007.⁶ The federal law standardised the documentation used to establish eligibility and limited the statu-

⁴See Law No. 12,933/2013, Article 1, https://www.planalto.gov.br/ccivil_03/_ato2011-2014/2013/lei/112933.htm.

⁵See Law No. 10,741/2003, Articles 1 and 23, https://www.planalto.gov.br/ccivil_03/leis/2003/l10.741.htm.

⁶See <https://www1.folha.uol.com.br/fsp/ilustrad/fq2904200707.htm> and <https://g1.globo.com/Noticias/Cinema/0,,MUL25098-7086,00.html>.

tory entitlement to 40% of the tickets available for each event.⁷ Elderly consumers' separate entitlement does not contain this limit.⁸

Eligibility is based on legally defined characteristics, and the law explicitly determined the student identification cards used to establish eligibility. It recognises cards issued by the National Student Union, the National Student Union of Secondary School Students, and the National Association of Postgraduate Students.⁹ Other eligible groups establish their entitlement through documentation of age, disability, or qualifying low-income status.

Cinemas are still free to offer voluntary discounts that are not determined by the statutory ratio, and the statutory benefit is not cumulative with these promotions. For example, Cinemark offers discounted tickets through its partnership with Banco Bradesco, while UCI and Kinoplex have similar partnerships with Banco Itaú. These arrangements create a separate price tier whose price is chosen voluntarily by each cinema.

2.2 Data and the observed price menu

Each cinema in Brazil must report ticket sales for every film session to the Brazilian National Film Agency (ANCINE). Through an FOI request, I obtained yearly aggregates of revenue and admissions by ticket category for all Brazilian municipalities reporting cinema activity between 2018 and 2025.¹⁰

Table 1 summarises national aggregates for the three paid ticket categories: full-price, statutory half-price, and promotional tickets. Total admissions also include complimentary tickets, which account for approximately 1.5–2.5% of admissions over the period. Statutory half-price tickets are the largest category in every year, accounting for more than 50% of admissions. The full-price share falls from 26.5%

⁷Decree No. 8,537/2015 sets out the reservation and disclosure rules.

⁸See Law No. 12,933/2013, Article 1, paragraph 10, and Decree No. 8,537/2015, Articles 9–11, https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/decreto/d8537.htm.

⁹Before the federal legislation, cinemas had to recognise cards issued by individual educational establishments.

¹⁰Legal restrictions prevent ANCINE from releasing the individual session records by ticket category. The category reporting requirement began in 2018, so comparable information is not available for earlier years.

in 2018 to 19.9% in 2025, while the promotional share rises from 17.6% to 24.2%. Thus, all three ticket categories are quantitatively relevant, and the decline in the full-price share is accompanied by an increase of the same magnitude in the promotional share.

Table 1. National aggregates, 2018–2025

Year	N municipalities	Tickets (M)	PMI (R\$)	Full-price (R\$)	Half-price (R\$)	Promo (R\$)	% Full-price	% Half-price	% Promo	Half/full
2018	405	164.4	14.89	21.82	12.80	12.36	26.5	54.2	17.6	0.587
2019	422	177.6	15.61	24.61	13.72	12.54	21.5	59.4	17.4	0.558
2020	426	39.9	15.88	25.44	14.00	13.00	20.0	61.0	17.3	0.550
2021	411	51.7	17.20	26.81	14.34	13.15	26.1	58.4	14.1	0.535
2022	423	97.9	18.80	29.38	16.39	13.92	23.4	59.6	15.4	0.558
2023	432	118.1	19.35	29.91	17.57	13.87	22.0	59.6	16.7	0.587
2024	447	129.2	19.58	30.96	18.14	14.42	19.9	57.4	20.8	0.586
2025	454	117.2	20.00	31.41	19.12	14.63	19.9	53.5	24.2	0.609

Note: N municipalities counts municipalities; M denotes millions. Full-price, half-price, and promotional prices are computed by dividing category revenue by category admissions, in nominal BRL. Omitted complimentary tickets account for around 1.5%–2.5% over the time period. PMI is total box-office revenue divided by all admissions.

ANCINE reports total revenue and admissions within each category. Thus, I compute the average ticket price for each category by dividing its revenue by its admissions. Between 2018 and 2025, the full price increases by 44.0%, close to cumulative inflation of 45.2%. The average promotional price increases by only 18.3%, implying a real decline of approximately 18.5%. Average box-office revenue per admission, reported in the PMI column, increases by 34.3% and falls by 7.5% in real terms.¹¹ Total admissions fall from 164.4 million in 2018 to 118.1 million in 2023 and 117.2 million in 2025, indicating that the sector had not returned to its pre-pandemic attendance level by the end of the period.

2.3 Aggregation and market structure

Notice that The half-price/full-price ratio in Table 1 does not reproduce the statutory ratio exactly. Each category average price aggregates across cinemas, films, screening times, formats, and locations. Eligible and full price consumers may purchase different combinations of screenings even when every individual cinema complies with the law. Thus, the average price ratio cannot be interpreted as a direct measure of compliance. In a conversation with a director of a large cinema chain, they confirmed that compliance to ratio is strict.

A separate ANCINE dataset containing session-level attendance provides information

¹¹The inflation adjustment uses annual IPCA changes published by the Brazilian Institute of Geography and Statistics.

about whether the statutory quota is likely to constrain half-price sales. Average occupancy was approximately 21% over 2017-2019. The proportion of sessions with attendance above 40% of room capacity was 14.5% in 2018, 11.2% in 2024, and 9.3% in 2025. Hence, in most sessions, total attendance was below the statutory quota. In addition, in a conversation with a director of a large cinema chain, they reported that, in the cinema industry, firms sell half-price tickets to eligible consumers even after discounted sales exceed 40% of session capacity.¹² Together, these facts suggest that every eligible consumer can purchase the statutory ticket for a cinema session.

Finally, Table 2 describes the structure of local cinema exhibition markets. Approximately 71% of municipalities with a cinema screen have a single active cinema complex.¹³

Table 2. Market structure by year

Year	Screens	Complexes	Municipalities	Single complex municipalities	Share (%)
2018	3281	784	396	282	71.2
2019	3447	822	414	292	70.5
2020	1825	446	272	202	74.3
2021	3238	748	402	290	72.1
2022	3378	793	419	300	71.6
2023	3434	823	430	305	70.9
2024	3499	856	443	317	71.6
2025	3516	868	447	317	70.9

3 Model

In this section, I present a model of relative price regulation. I use the institutional details provided in Section 2.1 to motivate four features of the model. Consistent with the prevalence of municipalities served by a single cinema complex, the model consists of a monopolist choosing a full posted price, while eligible consumers pay a fraction, fixed by regulation, of that price. I assume that any statutory quota is not

¹²This practice may differ in other cultural activities, such as music festivals, where organisers usually restrict discounted sales to the statutory quota.

¹³Table 1 includes every municipality in the revenue and admissions dataset, whereas Table 2 includes only municipalities matched to ANCINE's yearly cinema-complex data. The latter records the cinemas active in December of each year. A municipality in which a cinema closed before December may therefore appear in the revenue and admissions data but not in the cinema-complex dataset. Except for 2020, the coverage difference is small. Coverage is substantially lower in the pandemic year because many cinemas were not active in December.

binding, so all consumers entitled to the statutory discount can purchase the good. The firm may also offer a promotional ticket whose price is chosen voluntarily.

I first characterise how the regulated ratio affects the posted full price. Then, I introduce the voluntary promotional ticket and show how the result depends on whether its effective price can be chosen independently.

In terms of welfare, I compare the regulation with voluntary third-degree price discrimination. I then compare the ratio mandate with a cap that delivers the same eligible price. Finally, I consider extensions with fixed-percentage promotions, and consumers who qualify for both statutory and promotional discounts.

3.1 Setup and benchmark

¶ There is a unit mass of consumers. A mass $\alpha_H \in (0, 1)$ is eligible for the regulated discount, while the remaining $1 - \alpha_H$ is not. The subscripts F and H denote full-price and half-price consumers.¹⁴

Let $\varphi_j(p_j)$ denote the firm's per-capita operating profit from charging group $j \in \{F, H\}$ the price p_j . I assume that the relevant price range is an interval and that each φ_j is twice continuously differentiable and strictly concave on that interval, with a unique interior maximiser.

For the pricing results below, the only requirement is that the firm profit function is strictly concave and separable across groups. For example, with a common positive marginal cost c for both groups, one can have $\varphi_j(p_j) = (p_j - c)q_j(p_j)$.¹⁵

The firm chooses the posted full price p . Non-eligible consumers pay $p_F = p$, while eligible consumers pay

$$p_H = r p, \quad r \in (0, 1], \quad (1)$$

¹⁴Throughout the text, I use non-eligible and full-price consumers interchangeably, as well as eligible and half-price consumers.

¹⁵The pricing results also permit group-specific costs or fees, provided that the resulting per-capita profit functions remain strictly concave and separable across consumer groups.

where r is fixed by the regulator. For instance, in the Brazilian cinema case, this corresponds to $r = 1/2$. Under this regulated ratio, profit is

$$\pi_R(p) = (1 - \alpha_H)\varphi_F(p) + \alpha_H\varphi_H(rp). \quad (2)$$

Let $p_R^*(r, \alpha_H)$ denote the firm's optimal posted price under the regulated ratio r and eligible share α_H .

When doing so causes no ambiguity, I suppress the population share argument and write the optimal posted price as $p_R^*(r)$.

Without the regulation, in this type of cultural goods industry, firms usually incur in third-degree price discrimination. Thus, I compare the regulation to this scenario. In this case, the firm chooses the two group prices independently. For each group j , define $p_j^{PD} \equiv \arg \max_{p_j} \varphi_j(p_j) > 0$. These gives the unconstrained third-degree price discrimination prices p_F^{PD} and p_H^{PD} . The firm's preferred price ratio is then

$$r^* \equiv \frac{p_H^{PD}}{p_F^{PD}}. \quad (3)$$

The regulation binds whenever $r \neq r^*$. If $r < r^*$, the mandated discount is more generous than the firm would voluntarily offer; if $r > r^*$, it is less generous. When $r = r^*$, the regulation reproduces the voluntary price ratio.

3.2 Relative price regulation and the posted full price

Recall that p_H^{PD} is the profit maximising price for group H under unconstrained third-degree price discrimination. Under the regulated ratio, group H instead pays rp , where p is the posted full price. To ease the derivations, define the transformed eligible group profit function

$$\psi_r(p) \equiv \varphi_H(rp).$$

Because $r > 0$ and φ_H is strictly concave, ψ_r is also strictly concave, with a unique maximiser p_H^{PD}/r . At this posted price, group H pays

$$r \left(\frac{p_H^{PD}}{r} \right) = p_H^{PD},$$

which coincides with its profit maximising price under third-degree price discrimination.

The regulation therefore transforms the firm's problem into a common-price problem applied to $\varphi_F(p)$ and $\psi_r(p)$, with weights given by the two groups' population shares. The optimal posted price lies between the maximisers of the two component profits, p_F^{PD} and p_H^{PD}/r . These are the two groups' third-degree price discrimination benchmarks expressed in units of the posted full price. When $r = 1$, eligible consumers also pay p , so the problem reduces to the usual uniform pricing problem. More generally, the regulated ratio affects both the optimal posted price and its response to the eligible share, as the following proposition shows.

PROPOSITION 1 (Posted price under relative price regulation). *Suppose there is an interval of feasible posted prices containing p_F^{PD} and p_H^{PD}/r on which both $\varphi_F(p)$ and $\varphi_H(rp)$ are defined. Then:*

- (i) *If $r < r^*$, the mandated discount is more generous than the firm's preferred discount and*

$$p_F^{PD} < p_R^*(r, \alpha_H) < \frac{p_H^{PD}}{r}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} > 0.$$

- (ii) *If $r > r^*$, the mandated discount is less generous than the firm's preferred discount*

and

$$\frac{p_H^{PD}}{r} < p_R^*(r, \alpha_H) < p_F^{PD}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} < 0.$$

(iii) If $r = r^*$, the regulation reproduces the voluntary prices and

$$p_R^*(r, \alpha_H) = p_F^{PD} = \frac{p_H^{PD}}{r}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} = 0.$$

Consequently,

$$\text{sign} \left(p_R^*(r, \alpha_H) - p_F^{PD} \right) = \text{sign} \left(\frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} \right) = \text{sign}(r^* - r).$$

The intuition of this proposition is as follows. When $r < r^*$, the firm cannot retain both voluntary prices. At the voluntary full price, the mandated eligible price is below the eligible group's profit maximising level. Raising the posted price moves the eligible price towards that optimum but moves the full price away from its own optimum. Thus, the firm chooses a compromise with full-price consumers paying more and eligible consumers paying less than under voluntary discrimination.

A larger eligible share puts more weight on the profit coming from the eligible group, so the firm raises the posted price further. When $r > r^*$, the relative positions of the two voluntary benchmarks are reversed, so both price and eligible share comparative statics are reversed.

3.3 Relative price regulation and promotional discounts

As described in Section 2.1, the firm may also offer a voluntary promotional ticket alongside the regulated ticket. For example, the promotional group can represent consumers who have access to a bank card partnership. I incorporate this additional price by introducing a third consumer group. A mass α_P has access to a promotional

ticket, a mass α_H is eligible for the regulated price, and the remaining mass $\alpha_F = 1 - \alpha_H - \alpha_P$ pays the full price.

I assume initially that the three groups are mutually exclusive and have strictly positive masses. I extend the notation above so that φ_j , for every $j \in \{F, H, P\}$, is twice continuously differentiable and strictly concave, with a unique interior maximiser p_j^{PD} .

The firm chooses the posted full price p and the promotional discount $d \in [0, 1)$. The resulting price menu is

$$p_F = p, \quad p_H = r p, \quad p_P = p(1 - d).$$

For an interior promotional discount, choosing (p, d) is equivalent to choosing the posted and effective promotional prices (p, p_P) subject to $0 < p_P < p$. Thus, I can write the profit as

$$\pi_R(p, p_P) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(r p) + \alpha_P \varphi_P(p_P). \quad (4)$$

When the promotional price constraint is slack, p_P enters as separate term in the objective function, with the first two terms determining the posted price and the third determining the promotional price. Thus, only the relative sizes of the full price and eligible groups matter for the posted full price choice. Define the eligible share among these two groups as

$$\theta \equiv \frac{\alpha_H}{\alpha_F + \alpha_H} = \frac{\alpha_H}{1 - \alpha_P}. \quad (5)$$

Apart from the positive factor $1 - \alpha_P$, the part of profit that depends on p is

$$(1 - \theta)\varphi_F(p) + \theta\varphi_H(rp),$$

which is the problem in Proposition 1 with θ as the eligible share.

To compare this menu with third-degree price discrimination, define the voluntary promotional discount relative to the full price as

$$d_V^* \equiv 1 - \frac{p_P^{PD}}{p_F^{PD}}.$$

Under regulation, let $p_R^*(r, \theta)$ denote the optimal posted price and let

$$d_R^* \equiv 1 - \frac{p_P^{PD}}{p_R^*(r, \theta)}.$$

Applying Proposition 1 with θ in place of α_H yields the following corollary.

COROLLARY 1 (Promotional price separation benchmark). *Suppose there is an interval of feasible posted prices containing p_F^{PD} and p_H^{PD}/r on which both $\varphi_F(p)$ and $\varphi_H(rp)$ are defined. Suppose also that the voluntary and regulated promotional discounts are interior, so that $p_F^{PD} > p_P^{PD}$ and $p_R^*(r, \theta) > p_P^{PD}$. Then:*

(i) *The effective promotional price is*

$$p_P^R = p_P^{PD},$$

independently of the regulated ratio and the population shares.

(ii) *If $r < r^*$, relative price regulation raises the posted price and deepens the promotional discount:*

$$p_F^{PD} < p_R^*(r, \theta) < \frac{p_H^{PD}}{r}, \quad d_R^* > d_V^*,$$

and

$$\frac{\partial p_R^*(r, \theta)}{\partial \theta} > 0, \quad \frac{\partial d_R^*}{\partial \theta} > 0.$$

(iii) If $r > r^*$,

$$\frac{p_H^{PD}}{r} < p_R^*(r, \theta) < p_F^{PD}, \quad d_R^* < d_V^*,$$

and both comparative statics are negative. If $r = r^*$, the regulated and voluntary menus coincide and both derivatives are zero.

This corollary shows that the promotional tier does not change the effective price charged to group P . The reason is that the firm uses the discount d to keep that price at p_P^{PD} . However, it changes the discount expressed relative to the posted price. When regulation raises the posted price, the firm must advertise a larger percentage discount to preserve the same promotional price. Thus, a larger advertised markdown does not necessarily imply a lower transaction price.

The posted full price depends on the population shares through θ . Holding α_P fixed,

$$\frac{\partial \theta}{\partial \alpha_H} = \frac{1}{1 - \alpha_P} > 0,$$

so the comparative statics with respect to α_H have the same signs as those in the corollary.

These results rely on separable demand, fixed group membership, and an interior promotional discount. If the constraint $p_P < p$ binds, the firm cannot necessarily retain p_P^{PD} . I relax these assumptions in Section 3.6.

3.4 Welfare and the geometry of the ratio constraint

I now analyse the welfare effects of relative price regulation. When $r < r^*$, Proposition 1 shows that the regulation raises the posted full price while lowering the eligible price relative to voluntary discrimination. On the one hand, the lower eligible price attracts additional eligible consumers and transfers surplus to existing eligible consumers. On the other hand, the higher posted price deters some full-price consumers from purchasing the good. Thus, the welfare effects depend on the surplus gained from additional eligible consumers relative to the surplus lost from deterred full-price consumers.

As before, I compare the ratio mandate with voluntary third-degree price discrimination under the promotional price separation benchmark of Corollary 1. Group P pays p_P^{PD} in both regimes, so its welfare contribution cancels. Hence, I only need to compare groups F and H while allowing a positive promotional share.

The previous pricing exercises did not require specifying a demand function, but the welfare analysis does. I first consider linear demand and then generalise the analysis.

3.4.1 Linear demand as a weighted projection

Suppose that each group has linear per-capita demand

$$q_j(p_j) = a_j - b_j p_j, \quad a_j > 0, \quad b_j > 0,$$

and constant group-specific marginal cost $c_j < a_j/b_j$. I restrict attention to ratios for which both groups purchase positive quantities at the regulated optimum:

$$p_R^*(r) < \frac{a_F}{b_F}, \quad r p_R^*(r) < \frac{a_H}{b_H}.$$

The voluntary monopoly prices are

$$m_j \equiv p_j^{PD} = \frac{a_j + b_j c_j}{2b_j}.$$

Define

$$A \equiv \alpha_F b_F, \quad B \equiv \alpha_H b_H, \quad D \equiv A + Br^2, \quad z \equiv m_H - rm_F,$$

where the weights A and B combine each group's mass and demand slope. The gap z measures how far the voluntary eligible price lies above the eligible price required by the mandate at the voluntary full price. In particular, $z > 0$ exactly when $r < r^*$.

For vectors $x = (x_F, x_H)$ and $y = (y_F, y_H)$, define the weighted inner product and norm

$$\langle x, y \rangle_{A,B} \equiv Ax_F y_F + Bx_H y_H, \quad \|x\|_{A,B}^2 \equiv \langle x, x \rangle_{A,B}.$$

Let

$$m = (m_F, m_H), \quad c = (c_F, c_H),$$

and define the regulated ray

$$\mathcal{R}(r) \equiv \{(p_F, p_H) : p_H = r p_F, p_F \geq 0\}.$$

The regulation restricts the price vector to the ray from the origin with slope r .

If demand is linear, moving either price away from its voluntary optimum reduces profit quadratically. Thus, the firm chooses the point on the regulated ray closest to

m , with deviations weighted by A and B . Total welfare instead depends on distance from the efficient price vector c . Thus, the same constrained price vector is evaluated relative to m in the firm's objective function and relative to c in the welfare comparison.

PROPOSITION 2 (Projection geometry and welfare). *Under the promotional price separation benchmark, linear demand, constant group specific marginal costs, and interior solutions, the regulated price vector*

$$\mathbf{p}_R^*(r) = (p_R^*(r), r p_R^*(r)), \quad p_R^*(r) = \frac{Am_F + Brm_H}{D},$$

is the (A, B) -weighted orthogonal projection of m onto $\mathcal{R}(r)$. Let

$$u \equiv m - \mathbf{p}_R^*(r).$$

Total welfare and output satisfy

$$W_R(r) - W_V = \frac{1}{2} \|u\|_{A,B}^2 - \langle u, c \rangle_{A,B} \quad (6)$$

$$= \frac{AB}{2D} z [z - 2(c_H - rc_F)], \quad (7)$$

$$Q_R(r) - Q_V = \frac{AB(1-r)}{D} z. \quad (8)$$

Therefore:

- (i) For arbitrary group specific costs, relative price regulation lowers welfare if and only if z lies strictly between 0 and $2(c_H - rc_F)$, irrespective of the ordering of these two endpoints.
- (ii) If $c_F = c_H = 0$, relative price regulation yields weakly higher welfare than voluntary discrimination, with equality if and only if $z = 0$.
- (iii) If $c_F = c_H = c > 0$ and $0 < r < r^* < 1$, relative-price regulation lowers welfare if and

only if

$$0 < z < 2c(1 - r).$$

If $m_F > 2c$, define

$$r_{\text{low}} \equiv \frac{m_H - 2c}{m_F - 2c}.$$

The same condition is then

$$r_{\text{low}} < r < r^*.$$

If instead $m_H \leq 2c$, every binding ratio $r < r^*$ lowers welfare. Throughout this common cost reversal region, regulation raises total output while lowering welfare.

To illustrate this result, consider the following numerical example. Let the two groups have equal mass, demands $q_F(p) = 18 - p$, $q_H(p) = 10 - p$, and common marginal cost $c = 2$. The voluntary prices are $(m_F, m_H) = (10, 6)$. Using the Brazilian case as an example, a half-price mandate changes the price vector to $(10.4, 5.2)$.

In this exercise, aggregate full-price admissions fall by 0.2, while eligible admissions rise by 0.4, so total attendance rises by 0.2. The lost full-price admissions reduce surplus by 1.64, while the additional eligible admissions add 1.44, resulting in total welfare decreasing by 0.20. Higher attendance does not raise welfare because the lost full-price admissions generate more surplus per admission than the additional eligible admissions.

Equation (6) separates two components of the welfare change. The first,

$$\frac{1}{2} \|u\|_{A,B}^2,$$

is the zero cost welfare gain associated with moving away from the voluntary monopoly prices. The second,

$$\langle u, c \rangle_{A,B},$$

is the marginal cost correction. Because u is orthogonal to the regulated ray, only the component of the cost vector orthogonal to that ray affects the comparison.

For a fixed ratio, the welfare-loss region is a strip in voluntary price space, restricted to the maintained interior domain. Its boundaries are

$$m_H - rm_F = 0$$

and

$$m_H - rm_F = 2(c_H - rc_F).$$

The first boundary contains voluntary price menus that already satisfy the mandate. On the second boundary, the two components of the welfare change exactly offset.

With a common positive marginal cost and $r^* < 1$, suppose first that $m_H > 2c$. The threshold r_{low} then lies strictly between zero and r^* . A ratio just below r^* lowers welfare, whereas an admissible ratio below r_{low} raises it. The sign reversal applies over the range in which both groups remain served. If $m_H \leq 2c$, the threshold is not feasible and every binding ratio below r^* lowers welfare.

For a half-price mandate, the welfare-loss condition becomes

$$0 < m_H - \frac{1}{2}m_F < c.$$

When $m_F > 2c$, this is equivalently

$$r_{\text{low}} < \frac{1}{2} < r^*.$$

The same geometry identifies the ratio at which welfare is lowest. Continue with the common cost case and suppose that $r^* < 1$ and $m_H > 2c$. Define

$$\lambda \equiv \frac{\sqrt{Am_F^2 + Bm_H^2}}{\sqrt{A(m_F - 2c)^2 + B(m_H - 2c)^2}}, \quad r_w \equiv \frac{m_H + \lambda(m_H - 2c)}{m_F + \lambda(m_F - 2c)}.$$

Then

$$r_{\text{low}} < r_w < r^*.$$

The interior welfare formula falls with r up to r_w and rises thereafter. If the optimum at r_w keeps both groups served, this ratio attains the welfare minimum. On any closed interval throughout which the interior solution applies, welfare is maximised at an endpoint.

As the regulated ratio varies, $\mathbf{p}_R^*(r)$ traces an arc of the circle with diameter joining the origin and m in weighted coordinates. This is the Thales circle. The vectors

$$m - \mathbf{p}_R^*(r) \quad \text{and} \quad \mathbf{p}_R^*(r)$$

are orthogonal at every regulated optimum. Welfare falls with squared distance from c , so the reversal region is the part of the feasible Thales arc lying outside the circle centred at c and passing through m .

Geometrically, $\mathbf{p}_R^*(r_w)$ is the point where the line from the welfare centre c through the centre of the Thales circle, $m/2$, meets the feasible arc. An endpoint comparison

over all of $[0, 1]$ requires the interior conditions to hold throughout, including at the limiting case $r = 0$.

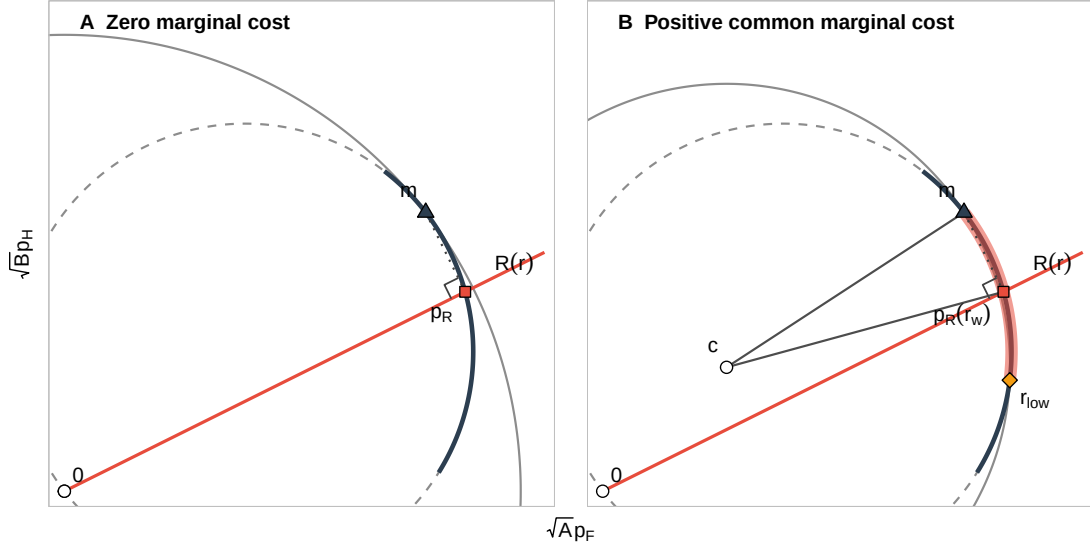


Figure 1. The geometry of relative price regulation in weighted price coordinates. The axes scale group prices by the square roots of $A = \alpha_F b_F$ and $B = \alpha_H b_H$. The vector m contains the voluntary monopoly prices, and c contains marginal costs. The dashed curve is the Thales circle with diameter $[0, m]$; its dark arc is traced by $\mathbf{p}_R^*(r)$ over the feasible ratios shown. The solid grey circle is centred at c and passes through m . At zero cost, the feasible arc lies inside the welfare boundary. With positive common cost, the red portion lies outside it and represents the welfare-loss region.

Equation (8) shows that regulation raises attendance throughout the common cost reversal region.

When $r = 1$ and the groups have a common marginal cost, equation (7) reduces to

$$W_R(1) - W_V = \frac{ABz^2}{2D} \geq 0.$$

The model therefore nests the familiar linear demand result that uniform monopoly pricing yields weakly higher total welfare than voluntary third-degree price discrimination (Schmalensee 1981; Varian 1985). Equation (8) also gives

$$Q_R(1) - Q_V = 0,$$

recovering the linear-demand output invariance under uniform pricing.

3.4.2 A local result under general demand

The linear model allows me to show a global welfare comparison over the range in which both groups remain served. I now show that the same mechanism arises locally under general demand. The comparison depends on both the curvature of marginal profit and the markup at the voluntary price.

The result is local because it evaluates welfare and output derivatives at the voluntary ratio. Away from that ratio, the regulated prices, and hence the relevant markups and demand curvature, change, so the derivatives may change sign. A global comparison would require additional shape restrictions over the entire price range, which is out of the scope of this paper.

For $j \in \{F, H\}$, let $q_j(p_j)$ be positive, twice continuously differentiable, and strictly decreasing near the voluntary monopoly price

$$m_j \equiv p_j^{PD}.$$

Let group j have constant marginal cost $c_j < m_j$, so that

$$\varphi_j(p_j) = (p_j - c_j)q_j(p_j),$$

and recall that

$$r^* = \frac{m_H}{m_F}.$$

Holding the group shares fixed, let $p_R^*(r)$ denote the firm's optimal posted price under ratio r . Define

$$\gamma_j \equiv -\varphi_j''(m_j) > 0, \quad L_j \equiv \frac{m_j - c_j}{m_j}.$$

The term γ_j measures how quickly marginal profit falls around the voluntary optimum, while L_j is the monopoly Lerner index.

Define the markup normalised curvature index

$$\chi_j \equiv \frac{(m_j - c_j)\gamma_j}{q_j(m_j)} = 2 - \frac{(m_j - c_j)^2 q_j''(m_j)}{q_j(m_j)}.$$

The index χ_j measures demand curvature after normalising by the markup and sales.

Let $W_R(r)$, $CS_R(r)$, and $Q_R(r)$ denote total welfare, consumer surplus, and output for groups F and H along the firm's constrained optimum. Let W_V , CS_V , and Q_V denote their counterparts under voluntary discrimination. Finally, define

$$K \equiv \alpha_F \gamma_F + \alpha_H (r^*)^2 \gamma_H.$$

PROPOSITION 3 (General demand near the voluntary ratio). *Under the conditions above, and for an admissible voluntary ratio $r^* \in (0, 1]$, using left derivatives when $r^* = 1$,*

$$\left. \frac{dW_R}{dr} \right|_{r=r^*} = \left. \frac{dCS_R}{dr} \right|_{r=r^*} \tag{9}$$

$$= \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{K} \left(\frac{\chi_H}{L_H} - \frac{\chi_F}{L_F} \right), \tag{10}$$

$$\left. \frac{dQ_R}{dr} \right|_{r=r^*} = \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{m_F K L_F L_H} \left(\chi_H - \frac{\chi_F}{r^*} \right). \tag{11}$$

Therefore, a sufficiently small reduction in the regulated ratio below r^ raises welfare and*

consumer surplus if

$$\frac{\chi_H}{L_H} < \frac{\chi_F}{L_F},$$

and raises total output if

$$\chi_H < \frac{\chi_F}{r^*}.$$

Proposition 3 shows that demand curvature and markups jointly determine the local welfare effect. With a common positive marginal cost, $r^* < 1$ implies $m_H < m_F$ and therefore $L_H < L_F$. If the two groups have equal curvature indices, lowering the ratio below r^* then lowers welfare. At the same time, equation (11) implies that the same reduction raises output.

Under linear demand, $\chi_F = \chi_H = 2$, so the local welfare comparison depends on the two Lerner indices alone. The general result therefore reproduces the local implication of Proposition 2.

With zero marginal cost, $L_F = L_H = 1$, and only the curvature comparison $\chi_H - \chi_F$ remains. A larger normalised convexity term

$$\frac{m_H^2 q_H''(m_H)}{q_H(m_H)}$$

reduces χ_H and favours a welfare improving reduction in the regulated ratio, echoing the curvature conditions in the price-discrimination literature (Aguirre et al. 2010; Cowan 2016).

The equality of the welfare and consumer surplus derivatives follows because r^* implements the voluntary profit maximising price vector. Maximised profit is therefore locally stationary with respect to r at the voluntary ratio.

3.5 Relative price regulation and an eligible price cap

To separate the effect of the discount from the effect of linking the two prices, I compare the ratio mandate with a cap fixing the price paid by eligible consumers. I set the cap equal to the eligible price produced by the ratio mandate:

$$\bar{p}_H = r p_R^*(r).$$

When this ceiling binds, the two instruments deliver the same eligible price.

The comparison is particularly simple when $r < r^*$. Under the ratio mandate, the full price is above its voluntary optimum. Under the cap, the firm can hold the eligible price fixed while choosing the full and promotional prices independently. Therefore, it therefore those prices to their voluntary optima. Eligible and promotional consumers face the same prices under the two instruments, while full-price consumers pay less and the firm earns more under the cap.

Let the superscripts V , R , and C denote voluntary discrimination, relative price regulation, and the matched cap. Let $W_C(r)$ denote welfare under the cap. Under the cap, the firm chooses p_F , p_H , and p_P separately, subject only to

$$p_H \leq \bar{p}_H.$$

As before, I omit the promotional group because the regulation does not affect its final price.

PROPOSITION 4 (Relative price regulation versus an eligible price cap). *Suppose that the assumptions of Corollary 1 hold, consumer surplus for each group is strictly decreasing in its price over the relevant range, and total welfare is the sum of profit and group consumer surplus. Then:*

(i) If $r < r^*$, the cap binds and produces

$$(p_F^C, p_H^C, p_P^C) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Relative-price regulation produces

$$(p_F^R, p_H^R, p_P^R) = (p_R^*(r), r p_R^*(r), p_P^{PD}), \quad p_R^*(r) > p_F^{PD}.$$

The cap therefore yields strictly higher profit, consumer surplus, and total welfare than relative-price regulation. It also restores the voluntary promotional discount:

$$d_C^* = d_V^* < d_R^*.$$

(ii) If $r = r^*$, the cap, relative price regulation, and voluntary discrimination produce the same price menu.

(iii) If $r > r^*$, the matched cap is slack and produces the voluntary price menu. It cannot reproduce the higher eligible price under relative price regulation, and the general profit function assumptions do not determine the welfare ranking.

Intuitively, both instruments give eligible consumers the same price, but the cap removes the link between that price and the rest of the menu. Full-price consumers and the firm are strictly better off, while eligible and promotional consumers face unchanged prices. The comparison is therefore also unchanged if additional non-negative welfare weight is placed on eligible consumers.

The cap comparison can also decompose the welfare effect of the ratio mandate. For any r , define the decoupled price menu

$$(p_F^D, p_H^D, p_P^D) = (p_F^{PD}, r p_R^*(r), p_P^{PD})$$

and let $W_D(r)$ denote its welfare. This menu combines the eligible price produced by the ratio mandate with the voluntary prices for the other groups. When $r < r^*$, it is

exactly the outcome of the matched cap, so

$$W_D(r) = W_C(r).$$

For every r , the decoupled menu separates the welfare effect of changing the eligible price from the cost of distorting the rest of the menu:

$$W_D(r) - W_V = \underbrace{W_D(r) - W_V}_{\text{eligible price component}} - \underbrace{W_D(r) - W_R(r)}_{\text{menu distortion}}. \quad (12)$$

Under general downward sloping demand and constant group specific marginal costs, the two components are

$$W_D(r) - W_V = \alpha_H \int_{r p_R^*(r)}^{m_H} (p - c_H) |q'_H(p)| dp, \quad (13)$$

$$W_D(r) - W_R(r) = \alpha_F \int_{m_F}^{p_R^*(r)} (p - c_F) |q'_F(p)| dp. \quad (14)$$

For $r < r^*$, the first component measures the surplus generated by the additional eligible admissions. It is positive if the eligible price remains above marginal cost. If the eligible price falls below marginal cost, the mandate can instead induce admissions whose resource cost exceeds willingness to pay. The second component measures the surplus lost when the higher full price deters consumers. The welfare effect of the ratio mandate is the first component less the second.

Under the linear demand notation of Proposition 2, the two components become

$$W_D(r) - W_V = \frac{ABz}{2D} [m_H + r p_R^*(r) - 2c_H], \quad (15)$$

$$W_D(r) - W_R(r) = \frac{ABr\alpha}{2D} [m_F + p_R^*(r) - 2c_F]. \quad (16)$$

Each component equals the change in the relevant group's aggregate admissions multiplied by its average price-cost margin over the adjustment. Subtracting the second component from the first reproduces equation (7).

For $\alpha > 0$, the reversal occurs when r times the full-price group's average markup exceeds the eligible group's average markup. With common positive marginal cost and $r^* < 1$, the voluntary eligible price has the smaller price-cost margin. Additional eligible admissions can therefore contribute less surplus per ticket than the full-price admissions lost. If this difference outweighs the increase in total attendance, welfare falls.

When $r > r^*$, the matched cap is slack and cannot reproduce the eligible price under the ratio mandate. Either welfare ranking is then possible. To illustrate, consider the two-group linear model with

$$\alpha_H = \frac{1}{2}, \quad b_F = b_H = 1, \quad r = 0.8.$$

First, let

$$(a_F, c_F) = (10, 0), \quad (a_H, c_H) = (6, 0).$$

Then $r^* = 0.6 < r$, the cap is slack, and

$$W_R(0.8) - W_C(0.8) = 0.152.$$

Alternatively, let

$$(a_F, c_F) = (12, 8), \quad (a_H, c_H) = (10, 0).$$

Then $r^* = 0.5 < r$, both regimes remain interior, and

$$W_R(0.8) - W_C(0.8) = -4.482.$$

Thus, when the cap is slack, relative price regulation can produce either higher or lower welfare than the cap. This ambiguity contrasts with the strict dominance of the cap when $r < r^*$.

The two instruments also differ in the form of the entitlement. When $r < r^*$, the cap produces the realised price ratio

$$\frac{p_H^C}{p_F^C} = \frac{r p_R^*(r)}{p_F^{PD}} > r.$$

Eligible consumers pay the same absolute price under both instruments, but the displayed percentage discount is smaller under the cap. Thus, a cap can reproduce the eligible price without reproducing a legal entitlement expressed as “half price.” The ratio makes the percentage entitlement directly observable while creating the menu distortion identified above.

3.6 Extensions to promotional pricing

The independent promotional price benchmark allows the firm to adjust the promotional discount separately in every market. I now consider two alternatives. First, the firm may fix the promotional percentage across markets, causing the effective promotional price to move with the posted price. Second, statutory and promotional eligibility may overlap, so that some consumers choose the lower of the two discounted prices.

3.6.1 Fixed-percentage promotions

Let

$$\rho \equiv 1 - d$$

and suppose that a chain fixes $\rho \in (0, 1)$ across its locations during a given period. The promotional price is then

$$p_P = \rho p,$$

and the posted price problem becomes

$$\pi_\rho(p) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(r p) + \alpha_P \varphi_P(\rho p). \quad (17)$$

The promotional term is now linked to the posted price, thus it enters the same transformed common price problem as the full-price and eligible terms.

PROPOSITION 5 (Fixed-percentage promotions). *Suppose the three profit functions in Equation (17) are strictly concave, their transformed maximisers lie in the feasible interval, and all three group shares are positive. Then π_ρ has a unique maximiser p_ρ^* lying between*

$$p_F^{PD}, \quad \frac{p_H^{PD}}{r}, \quad \frac{p_P^{PD}}{\rho}.$$

For any market shifter v that leaves ρ fixed,

$$\frac{\partial \log p_P}{\partial v} = \frac{\partial \log p_\rho^*}{\partial v}.$$

Holding α_P fixed and increasing α_H at the expense of α_F gives

$$\frac{\partial p_\rho^*}{\partial \alpha_H} = - \frac{r \phi'_H(r p_\rho^*) - \phi'_F(p_\rho^*)}{\pi''_\rho(p_\rho^*)}.$$

Increasing α_P at the expense of α_F gives the analogous expression with

$$\rho \phi'_P(\rho p_\rho^*)$$

in place of

$$r \phi'_H(r p_\rho^*).$$

In particular, if

$$p_F^{PD} \leq \frac{p_P^{PD}}{\rho} \leq \frac{p_H^{PD}}{r} \quad \text{and} \quad p_F^{PD} < \frac{p_H^{PD}}{r},$$

then

$$\frac{\partial p_\rho^*}{\partial \alpha_H} > 0.$$

A fixed-percentage promotional price is therefore a third ratio linked term in the transformed common price problem. Because $p_P = \rho p$, the promotional price moves one-for-one in logs with the posted price. In this setting, the promotional group also affects the optimal posted price.

Under the ordering in the proposition, the promotional benchmark lies between the voluntary full price benchmark and the posted price that would implement the eligible group's voluntary optimum. This ordering preserves the positive comparative static of the posted price with respect to the eligible share. It does not, by itself, determine the effect of the promotional share, which depends on the relative

marginal profits of the promotional and full price groups.

The observed promotional category can therefore combine two forms of pricing. Some promotional prices may follow the separation benchmark and remain independent of the posted price. Others may be set as a fixed percentage of the posted price and move with it.

3.6.2 Overlapping eligibility

The separation benchmark also assumes that statutory and promotional eligibility are disjoint. In practice, a consumer may qualify for both discounts and pay

$$\min\{r p, p_P\}.$$

Let the partition the population into four mutually exclusive groups with masses

$$\alpha_F, \quad \alpha_{H \setminus P}, \quad \alpha_{P \setminus H}, \quad \alpha_B,$$

which sum to one. The groups $H \setminus P$ and $P \setminus H$ contain consumers eligible only for the statutory discount and only for the promotion. Group B contains consumers eligible for both.

The firm's objective is kinked at

$$p_P = r p.$$

Immediately below the kink, changing p_P affects both promotional-only and dual-eligible consumers. Immediately above it, changing p_P affects only promotional-only consumers.

Let φ_B denote per-capita profit from a dual-eligible consumer. At a given posted price p , setting $p_P = r p$ is locally optimal with respect to p_P when

$$\alpha_{P \setminus H} \varphi'_P(r p) + \alpha_B \varphi'_B(r p) \geq 0$$

and

$$\alpha_{P \setminus H} \varphi'_P(r p) \leq 0.$$

Define the pooled promotional optimum

$$p_{P \cup B}^* \equiv \arg \max_x \{ \alpha_{P \setminus H} \varphi_P(x) + \alpha_B \varphi_B(x) \}.$$

Strict concavity makes the two inequalities equivalent to

$$p_P^{PD} \leq r p \leq p_{P \cup B}^*.$$

The promotional price therefore coincides with the statutory price whenever the mandated eligible price lies between the promotional-only optimum and the pooled optimum for promotional-only and dual-eligible consumers. This interval is non-empty only when pooling the dual-eligible group weakly raises the preferred promotional price.

When the kink binds, the part of profit that depends on the posted price is

$$\alpha_F \varphi_F(p) + \alpha_{H \setminus P} \varphi_H(r p) + \alpha_{P \setminus H} \varphi_P(r p) + \alpha_B \varphi_B(r p).$$

Define the total mass whose price is linked to the statutory ratio as

$$\beta \equiv \alpha_{H \setminus P} + \alpha_{P \setminus H} + \alpha_B$$

and define its average per-capita profit function as

$$\bar{\varphi}_E(x) \equiv \frac{\alpha_{H \setminus P} \varphi_H(x) + \alpha_{P \setminus H} \varphi_P(x) + \alpha_B \varphi_B(x)}{\beta}.$$

The posted-price-dependent part of profit can then be written as

$$\alpha_F \varphi_F(p) + \beta \bar{\varphi}_E(rp).$$

Holding the composition of the ratio linked group fixed, Proposition 1 applies with β in place of α_H and $\bar{\varphi}_E$ in place of φ_H . Within a regime in which the kink remains optimal, overlapping eligibility therefore expands the mass whose price is tied to the posted price. An increase in that mass raises the posted price when the pooled group's preferred ratio exceeds r .

4 Conclusion

The present paper studies relative price regulation. This type of regulation arises when a rule fixes the price paid by eligible consumers as a fraction of the posted full price paid by non-eligible consumers, whilst allowing the firm to choose the price level and any other prices in its menu.

I show that when the regulation requires a larger percentage discount than the firm would voluntarily offer, the firm raises the posted full price and lowers the eligible price compared to voluntary discrimination. This response becomes stronger as the share of eligible consumers increases, shifting more of the cost of the regulation onto consumers who pay the full price. Additionally, if the firm can offer an independently

chosen promotional price, that price remains unchanged even though the higher full price makes the advertised percentage discount larger.

The welfare effect of this regulation depends on how consumption changes across the two groups. The lower regulated price increases consumption among eligible consumers, but the higher full price reduces consumption among non-eligible ones. Thus, total output can rise whilst total welfare falls if the purchases deterred among full-price consumers would have generated more surplus than the additional purchases by eligible consumers. With a linear demand function and a common positive marginal cost, I derive the exact condition for this result. For example, with a half-price mandate, and provided that both groups remain served, welfare falls when the voluntary eligible price is above half the voluntary full price, but the difference is smaller than marginal cost. With a general demand function, the local result also depends on demand curvature and the monopoly markups in the two groups.

To separate the effect of the discount from the effect of linking the two prices, I compare the ratio mandate with a cap fixing the price paid by eligible consumers. The cap removes the firm's incentive to raise the posted full price, returning it to its voluntary level, while the eligible price remains unchanged. Hence, non-eligible consumers pay less, eligible consumers pay the same price, and the firm earns a higher profit. Consumer surplus and total welfare are also greater under the cap. This comparison points to the central trade-off of the regulation. It leaves the price level to the firm, but doing so lets the firm shift part of the adjustment onto the unregulated reference price.

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Appendix A. Proofs and additional results

Proof of Proposition 1. For this proof, write $p_R^* = p_R^*(r, \alpha_H)$. The first-order condition for the regulated problem is

$$G(p, \alpha_H) \equiv (1 - \alpha_H)\phi_F'(p) + \alpha_H r \phi_H'(r p) = 0.$$

Because

$$G_p(p, \alpha_H) = (1 - \alpha_H)\phi_F''(p) + \alpha_H r^2 \phi_H''(r p) < 0,$$

the regulated objective is strictly concave, so any interior solution is unique.

At the voluntary full price,

$$G(p_F^{PD}, \alpha_H) = \alpha_H r \phi_H'(r p_F^{PD}),$$

because $\phi_F'(p_F^{PD}) = 0$. Strict concavity of ϕ_H implies

$$\text{sign}\left[\phi_H'(r p_F^{PD})\right] = \text{sign}(p_H^{PD} - r p_F^{PD}) = \text{sign}(r^* - r).$$

At the price that implements the eligible group's voluntary price,

$$G\left(\frac{p_H^{PD}}{r}, \alpha_H\right) = (1 - \alpha_H)\phi_F'\left(\frac{p_H^{PD}}{r}\right),$$

because $\phi_H'(p_H^{PD}) = 0$.

If $r < r^*$, then $p_F^{PD} < p_H^{PD}/r$. Consequently,

$$G(p_F^{PD}, \alpha_H) > 0 \quad \text{and} \quad G\left(\frac{p_H^{PD}}{r}, \alpha_H\right) < 0.$$

Continuity gives a zero between the two endpoints, and because G is strictly decreas-

ing in p , that zero is unique. It satisfies

$$p_F^{PD} < p_R^* < \frac{p_H^{PD}}{r}.$$

If $r > r^*$, the endpoint inequalities and the signs of G are reversed, giving

$$\frac{p_H^{PD}}{r} < p_R^* < p_F^{PD}.$$

If $r = r^*$, the two endpoints coincide and $p_R^* = p_F^{PD} = p_H^{PD}/r$.

To obtain the comparative static, differentiate the first-order condition implicitly:

$$\frac{\partial p_R^*}{\partial \alpha_H} = -\frac{G_{\alpha_H}}{G_p}.$$

At the optimum,

$$G_{\alpha_H} = -\varphi'_F(p_R^*) + r\varphi'_H(rp_R^*) = -\frac{1}{\alpha_H}\varphi'_F(p_R^*),$$

where the second equality uses the first-order condition. Therefore,

$$\frac{\partial p_R^*}{\partial \alpha_H} = \frac{\varphi'_F(p_R^*)}{\alpha_H [(1 - \alpha_H)\varphi''_F(p_R^*) + \alpha_H r^2 \varphi''_H(rp_R^*)]}.$$

The denominator is negative. Strict concavity of φ_F implies that $\varphi'_F(p_R^*) < 0$ when $p_R^* > p_F^{PD}$ and $\varphi'_F(p_R^*) > 0$ when $p_R^* < p_F^{PD}$. Hence

$$\text{sign}\left(\frac{\partial p_R^*}{\partial \alpha_H}\right) = \text{sign}(p_R^* - p_F^{PD}) = \text{sign}(r^* - r).$$

□

Proof of Corollary 1. For this proof, write $p_R^* = p_R^*(r, \theta)$. For an interior promotional discount, choosing (p, d) is equivalent to choosing (p, p_P) , where $p_P = p(1 - d)$, subject to $0 < p_P < p$. By equation (4), profit is

$$\pi_R(p, p_P) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(rp) + \alpha_P \varphi_P(p_P).$$

Because the objective is separable in p and p_p , and φ_p has the unique interior maximiser p_p^{PD} , the optimal effective promotional price is

$$p_p^R = p_p^{PD}.$$

The maintained condition $p_R^* > p_p^{PD}$ ensures that the constraint $p_p < p$ is slack at the regulated solution. Similarly, $p_F^{PD} > p_p^{PD}$ ensures an interior promotional discount under voluntary discrimination. The effective promotional price is therefore independent of the regulated ratio and the population shares.

Using equation (5), the part of profit that depends on the posted price can be written as

$$(1 - \alpha_p) [(1 - \theta)\varphi_F(p) + \theta\varphi_H(rp)].$$

The positive factor $1 - \alpha_p$ does not affect the optimum. Proposition 1 therefore applies directly with θ in place of α_H . Hence

$$\text{sign}(p_R^* - p_F^{PD}) = \text{sign}\left(\frac{\partial p_R^*}{\partial \theta}\right) = \text{sign}(r^* - r),$$

with p_R^* lying strictly between p_F^{PD} and p_H^{PD}/r whenever $r \neq r^*$.

Finally,

$$d_R^* - d_V^* = p_p^{PD} \left(\frac{1}{p_F^{PD}} - \frac{1}{p_R^*} \right) = \frac{p_p^{PD}(p_R^* - p_F^{PD})}{p_F^{PD} p_R^*},$$

so the discount comparison has the same sign as the posted-price comparison.

Moreover,

$$\frac{\partial d_R^*}{\partial \theta} = \frac{p_p^{PD}}{(p_R^*)^2} \frac{\partial p_R^*}{\partial \theta}.$$

Thus the discount comparative static also has the sign of $r^* - r$. When $r = r^*$, the posted prices and discounts coincide and both derivatives are zero. $\square$

A.1 Linear-demand pricing formulas

For closed-form pricing formulas, specialise to linear per-capita demand and zero marginal cost:

$$q_j(p_j) = a_j - b_j p_j, \quad a_j > 0, \quad b_j > 0.$$

The parameters (a_j, b_j) describe per-capita demand and do not incorporate group size. In this case, $\varphi_j(p_j) = p_j q_j(p_j)$, and voluntary discrimination gives

$$p_F^{PD} = \frac{a_F}{2b_F}, \quad p_H^{PD} = \frac{a_H}{2b_H}, \quad (\text{A.1})$$

so that

$$r^* = \frac{a_H b_F}{a_F b_H}.$$

The optimal regulated posted price in the two-group model is

$$p_R^*(r, \alpha_H) = \frac{(1 - \alpha_H)a_F + r\alpha_H a_H}{2[(1 - \alpha_H)b_F + r^2\alpha_H b_H]}. \quad (\text{A.2})$$

An interior solution requires $p_R^*(r, \alpha_H) < a_F/b_F$ and $r p_R^*(r, \alpha_H) < a_H/b_H$. At $r = 1/2$, equation (A.2) becomes

$$p_{HL}^*(\alpha_H) = \frac{2(1 - \alpha_H)a_F + \alpha_H a_H}{4(1 - \alpha_H)b_F + \alpha_H b_H}. \quad (\text{A.3})$$

If the two groups have identical per-capita demand, so that $a_F = a_H = a$ and $b_F = b_H = b$, then

$$p_{HL}^*(\alpha_H) = \frac{a(2 - \alpha_H)}{b(4 - 3\alpha_H)}, \quad \frac{\partial p_{HL}^*}{\partial \alpha_H} = \frac{2a}{b(4 - 3\alpha_H)^2} > 0.$$

Taking the log derivative gives

$$\frac{\partial \log p_{HL}^*}{\partial \alpha_H} = \frac{2}{(4 - 3\alpha_H)(2 - \alpha_H)}, \quad (\text{A.4})$$

which is positive and increasing in $\alpha_H \in (0, 1)$.

With the promotional group included, the linear model gives

$$p_P^{PD} = \frac{a_P}{2b_P}, \quad (\text{A.5})$$

and

$$p_R^*(r, \theta) = \frac{\alpha_F a_F + r \alpha_H a_H}{2(\alpha_F b_F + r^2 \alpha_H b_H)}, \quad d_R^* = 1 - \frac{p_P^{PD}}{p_R^*(r, \theta)}. \quad (\text{A.6})$$

These formulas illustrate the separation benchmark in Corollary 1.

Proof of Proposition 2. Under the separation benchmark, group P is charged p_P^{PD} in both regimes, so its contribution to welfare cancels. For groups F and H , linear demand with constant marginal cost implies that per-capita profit can be written as

$$\varphi_j(p_j) = \varphi_j(m_j) - b_j(p_j - m_j)^2.$$

Thus aggregate profit is a constant minus $\|p - m\|_{A,B}^2$. Minimising this squared distance over $\mathcal{R}(r)$ gives

$$p_R^*(r) = \frac{Am_F + Brm_H}{D}.$$

Equivalently, define the residual $u \equiv m - \mathbf{p}_R^*(r)$ by

$$u_F \equiv m_F - p_R^*(r) = -\frac{Brz}{D}, \quad u_H \equiv m_H - r p_R^*(r) = \frac{Az}{D}.$$

It satisfies

$$\langle u, (1, r) \rangle_{A,B} = Au_F + Bru_H = 0,$$

so $\mathbf{p}_R^*(r)$ is the weighted orthogonal projection of m onto the regulated ray.

Linear-demand total welfare can be written as a constant minus $\|p - c\|_{A,B}^2/2$. Hence

$$W_R(r) - W_V = \frac{1}{2} \left(\|m - c\|_{A,B}^2 - \|\mathbf{p}_R^*(r) - c\|_{A,B}^2 \right).$$

Expanding this difference and using the orthogonality condition gives

$$\begin{aligned} W_R(r) - W_V &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) + Au_F(p_R^*(r) - c_F) + Bu_H(r p_R^*(r) - c_H) \\ &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) + p_R^*(r)(Au_F + Bru_H) - (Ac_F u_F + Bc_H u_H) \\ &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) - (Ac_F u_F + Bc_H u_H) \\ &= \frac{AB}{2D} z^2 - \frac{AB}{D} z(c_H - rc_F) \\ &= \frac{AB}{2D} z \left[z - 2(c_H - rc_F) \right], \end{aligned}$$

which proves equations (6) and (7).

For total output, linear demand gives

$$\begin{aligned} Q_R(r) - Q_V &= A(m_F - p_R^*(r)) + B(m_H - r p_R^*(r)) \\ &= Au_F + Bu_H \\ &= \frac{AB(1-r)}{D} z. \end{aligned}$$

This proves equation (8). For arbitrary c_F and c_H ,

$$W_R(r) < W_V \iff z[z - 2(c_H - rc_F)] < 0.$$

The inequality holds exactly when z lies strictly between the two roots 0 and $2(c_H - rc_F)$.

If $c_F = c_H = 0$, the expression reduces to $ABz^2/(2D)$, which is non-negative and equals zero if and only if $z = 0$. If $c_F = c_H = c > 0$ and $0 < r < r^* < 1$, then $z > 0$ and $c_H - rc_F = c(1-r) > 0$. The welfare difference is negative if and only if $0 < z < 2c(1-r)$, and equation (8) is strictly positive throughout this interval.

When $m_F > 2c$, substituting $z = m_H - rm_F$ into the upper inequality gives

$$m_H - rm_F < 2c(1 - r) \iff r > \frac{m_H - 2c}{m_F - 2c} \equiv r_{\text{low}}.$$

If $m_H > 2c$, then $0 < r_{\text{low}} < r^*$, which establishes the non-monotonicity result. If $m_H \leq 2c$, define $g(r) \equiv 2c(1 - r) - z = 2c - m_H + r(m_F - 2c)$. This affine function is weakly positive at both endpoints of $[0, r^*]$: $g(0) = 2c - m_H \geq 0$ and $g(r^*) = 2c(1 - r^*) > 0$. Hence it is positive for every $r \in (0, r^*)$, apart from a possible equality at the excluded endpoint $r = 0$. Finally, setting $r = 1/2$ in $0 < z < 2c(1 - r)$ gives $0 < m_H - m_F/2 < c$.

It remains to locate the welfare minimum. In weighted coordinates, write

$$\tilde{m} = (\sqrt{A}m_F, \sqrt{B}m_H), \quad \tilde{c} = (\sqrt{A}c, \sqrt{B}c).$$

The Thales circle has centre $\tilde{m}/2$ and radius $\|\tilde{m}\|/2$. Its point farthest from $\tilde{c}$ is

$$\frac{\tilde{m}}{2} + \frac{\|\tilde{m}\|}{2} \frac{\tilde{m} - 2\tilde{c}}{\|\tilde{m} - 2\tilde{c}\|}.$$

When $m_H > 2c$ and $r^* < 1$, both coordinates of $\tilde{m} - 2\tilde{c}$ are positive, so this point belongs to the arc defined by ratios in $(0, 1)$. Its interpretation as a welfare minimum also requires positive demand from both groups at the corresponding prices. Transforming back to price coordinates gives the ratio r_w stated in the text. It lies strictly between $r_{\text{low}} = (m_H - 2c)/(m_F - 2c)$ and $r^* = m_H/m_F$ by the strict mediant inequality. Squared distance from $\tilde{c}$ increases along the feasible arc up to this point and decreases thereafter. Since welfare is a constant minus one-half of that squared distance, it falls up to r_w and then rises, proving the remaining claims. $\square$

Proof of Proposition 3. At $r = r^*$, the constrained and voluntary price vectors coincide: $p_R^*(r^*) = m_F$ and $r^* p_R^*(r^*) = m_H$. Differentiating the firm's first-order condition

$$\alpha_F \phi'_F(p_R^*(r)) + \alpha_H r \phi'_H(r p_R^*(r)) = 0$$

with respect to r and evaluating at r^* gives

$$p_R^{*'}(r^*) = -\frac{\alpha_H r^* m_F \gamma_H}{K}, \quad \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} = \frac{\alpha_F m_F \gamma_F}{K}.$$

Group j 's total-surplus derivative with respect to its price is $(p_j - c_j)q'_j(p_j)$. The monopoly first-order condition implies

$$q_j(m_j) + (m_j - c_j)q'_j(m_j) = 0,$$

so $W'_j(m_j) = -q_j(m_j)$. Applying the chain rule and substituting the two price derivatives therefore yields

$$\begin{aligned} \left. \frac{dW_R}{dr} \right|_{r=r^*} &= -\alpha_F q_F(m_F) p_R^{*'}(r^*) - \alpha_H q_H(m_H) \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} \\ &= \frac{\alpha_F \alpha_H m_F}{K} [r^* \gamma_H q_F(m_F) - \gamma_F q_H(m_H)] \\ &= \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{K} \left(\frac{\chi_H}{L_H} - \frac{\chi_F}{L_F} \right). \end{aligned}$$

The last equality uses $m_H = r^* m_F$ and $\gamma_j = \chi_j q_j(m_j)/(L_j m_j)$.

Let $\pi_R^*(r)$ denote maximised profit under the ratio. The envelope theorem gives

$$\left. \frac{d\pi_R^*}{dr} \right|_{r=r^*} = \alpha_H m_F \phi'_H(m_H) = 0.$$

Since $W_R(r) = CS_R(r) + \pi_R^*(r)$, the welfare and consumer-surplus derivatives coincide at r^* .

The same steps for total output give

$$\begin{aligned} \left. \frac{dQ_R}{dr} \right|_{r=r^*} &= \alpha_F q'_F(m_F) p_R^{*'}(r^*) + \alpha_H q'_H(m_H) \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} \\ &= \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{m_F K L_F L_H} \left(\chi_H - \frac{\chi_F}{r^*} \right). \end{aligned}$$

Because $W_R(r^*) = W_V$ and $Q_R(r^*) = Q_V$, continuity gives the stated local comparisons for a sufficiently small reduction in r . $\square$

Proof of Proposition 4. Under the eligible-price cap, profit is

$$\pi_C(p_F, p_H, p_P) = \alpha_F \varphi_F(p_F) + \alpha_H \varphi_H(p_H) + \alpha_P \varphi_P(p_P), \quad p_H \leq \bar{p}_H.$$

The objective is separable. The unique unconstrained maximisers are therefore p_F^{PD} , p_H^{PD} , and p_P^{PD} .

For any r , define the decoupled menu

$$(p_F^D, p_H^D, p_P^D) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Adding and subtracting its welfare gives

$$W_R(r) - W_V = (W_D(r) - W_V) - (W_D(r) - W_R(r)).$$

Since group j 's total-surplus derivative is $(p - c_j)q'_j(p)$, and the two comparisons change only the eligible and full prices, respectively,

$$\begin{aligned} W_D(r) - W_V &= \alpha_H \int_{r p_R^*(r)}^{m_H} (p - c_H) |q'_H(p)| dp, \\ W_D(r) - W_R(r) &= \alpha_F \int_{m_F}^{p_R^*(r)} (p - c_F) |q'_F(p)| dp, \end{aligned}$$

which proves equations (13) and (14) for every r . Under linear demand, $m_H - r p_R^*(r) = Az/D$ and $p_R^*(r) - m_F = Brz/D$. Evaluating the two integrals then gives equations (15) and (16).

Suppose first that $r < r^*$. Corollary 1 implies

$$p_F^{PD} < p_R^*(r) < \frac{p_H^{PD}}{r}.$$

Consequently, $\bar{p}_H = r p_R^*(r) < p_H^{PD}$, so the cap binds. The optimal menu under the cap is

$$(p_F^C, p_H^C, p_P^C) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Under relative-price regulation, the menu is

$$(p_F^R, p_H^R, p_P^R) = (p_R^*(r), r p_R^*(r), p_F^{PD}).$$

The two regimes give the eligible and promotional groups identical prices. For the full-price group, strict concavity and the uniqueness of p_F^{PD} give

$$\pi_C - \pi_R = \alpha_F \left[\varphi_F(p_F^{PD}) - \varphi_F(p_R^*(r)) \right] > 0.$$

Because $p_F^{PD} < p_R^*(r)$ and full-price consumer surplus is strictly decreasing in price, aggregate consumer surplus is also strictly higher under the cap. Total welfare, defined as the sum of profit and consumer surplus, is therefore strictly higher.

Moreover,

$$d_C^* = 1 - \frac{p_P^{PD}}{p_F^{PD}} = d_V^*, \quad d_R^* = 1 - \frac{p_P^{PD}}{p_R^*(r)}.$$

Since $p_R^*(r) > p_F^{PD}$, it follows that $d_R^* > d_C^* = d_V^*$.

If $r = r^*$, Corollary 1 gives $p_R^*(r) = p_F^{PD}$ and $r p_R^*(r) = p_H^{PD}$. The cap is weakly binding and all three regimes produce the voluntary menu.

Finally, if $r > r^*$, Corollary 1 gives

$$\frac{p_H^{PD}}{r} < p_R^*(r) < p_F^{PD}.$$

Hence $\bar{p}_H = r p_R^*(r) > p_H^{PD}$, so the cap is slack and the firm chooses the voluntary menu $(p_F^{PD}, p_H^{PD}, p_P^{PD})$. The welfare comparison is therefore the comparison between relative-price regulation and voluntary discrimination. The two linear-demand examples following the proposition satisfy the maintained assumptions and produce opposite welfare rankings, establishing that the general assumptions do not determine the sign. $\square$

Proof of Proposition 5. Define the transformed profit functions

$$\psi_H(p) \equiv \varphi_H(rp), \quad \psi_P(p) \equiv \varphi_P(\rho p).$$

They are strictly concave and have respective maximisers p_H^{PD}/r and p_P^{PD}/ρ . Hence π_ρ is strictly concave. At the smallest of the three transformed maximisers every component has a weakly non-negative derivative, while at the largest every component has a weakly non-positive derivative. The unique zero of π'_ρ therefore lies between them.

Because $p_P = \rho p_\rho^*$ and ρ is fixed, $d \log p_P = d \log p_\rho^*$. For the share comparative statics, use $\alpha_F = 1 - \alpha_H - \alpha_P$ and differentiate the first-order condition

$$\alpha_F \varphi'_F(p) + \alpha_H r \varphi'_H(rp) + \alpha_P \rho \varphi'_P(\rho p) = 0.$$

Holding α_P fixed gives

$$\frac{\partial p_\rho^*}{\partial \alpha_H} = - \frac{r \varphi'_H(rp_\rho^*) - \varphi'_F(p_\rho^*)}{\pi''_\rho(p_\rho^*)}.$$

Holding α_H fixed and varying α_P yields the stated analogous formula. The denominator is strictly negative in both expressions. Under the additional ordering in the proposition, the bracketing result gives $p_F^{PD} < p_\rho^* < p_H^{PD}/r$. Strict concavity therefore implies $\varphi'_F(p_\rho^*) < 0$ and $\varphi'_H(rp_\rho^*) > 0$. The numerator in the displayed comparative static is strictly positive, so $\partial p_\rho^*/\partial \alpha_H > 0$. $\square$